TrES-3b

R-0593

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-3_230822` · created 2026-07-24T23:51:01+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460178.6915 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.188 +/- 0.057
Transit depth (Rp/Rs)^2	3.54 +/- 2.14 [%]
Orbital Inclination (inc)	86.77 +/- 2.82
Airmass coefficient 1 (a1)	0.61 +/- 0.3
Airmass coefficient 2 (a2)	0.43 +/- 1.0
Scatter in the residuals of the lightcurve fit is	51.01 %
Best Comparison Star	#2 - [302, 370]
Optimal Aperture	2.26
Optimal Annulus	7.32
Transit Duration (day)	0.079 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.188 ± 0.057 vs 0.16309 (deviation 0.42σ)
Duration fit vs published	0.079 d vs 0.0567 d (deviation 1.95σ)
Mid-time O–C	-4.2 min
Depth SNR	3.2
Residual scatter	51.01 %

Light curve

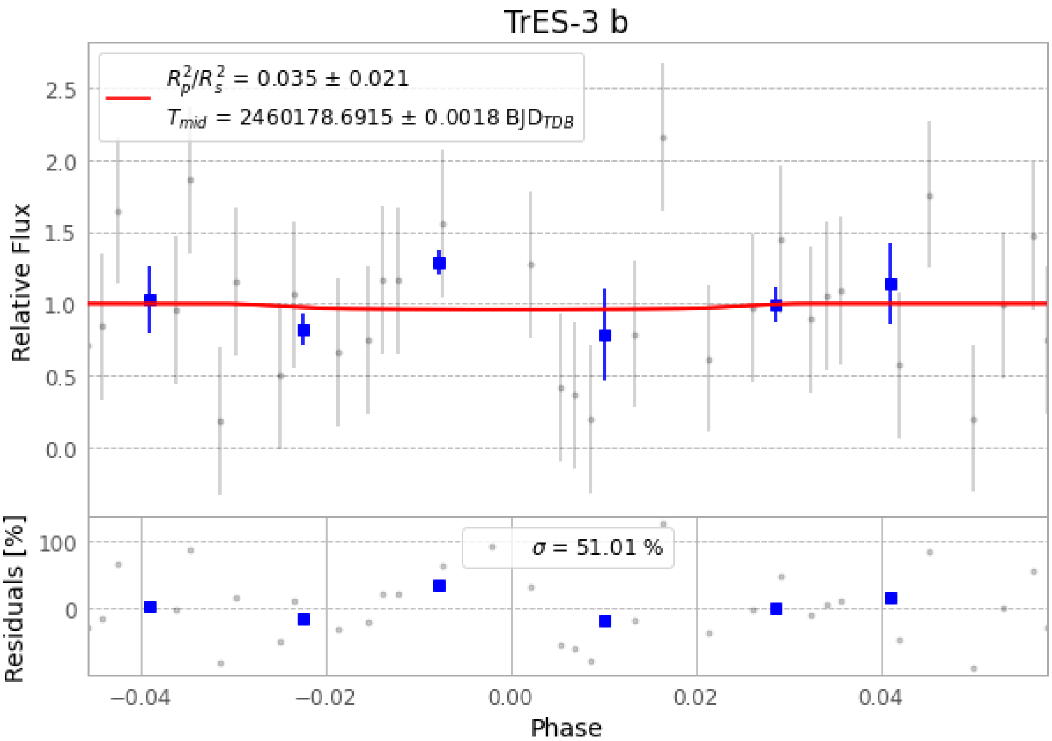

Figure 1 — FinalLightCurve_TrES-3 b_2023-08-21.png (SHA-256 c7475d00b608618a…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [581, 388], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 d33fbe4890279c25…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494`

Data provenance

68 FITS frames (45 MB), `TRES-3230822030016.FITS` → `TRES-3230822062115.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-230822.log (33fbd2dd3099…), FinalLightCurve_TrES-3 b_2023-08-21.png (c7475d00b608…), FinalLightCurve_TrES-3 b_2023-08-21.csv (a10e5b5268a8…)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)